

Lab Title:- : PC SYSTEM FUNDAMENTALS, DISK SECTORS AND PARTITION ANALYSIS

Student Name:- BADAMASI MUNTAKA MUHAMMAD

Student ID:- 2025FWSD11563

Course Name:- BASIC COMPUTER SKILLS FOR DIGITAL FORENSICS

Instructor Name:- COMMANDANT AMINU IDRIS

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Executive summary

The lab covers a practical understanding of some basic system hardware and software information of the window/linux operating system which involves understanding and calculating of the disk and partition sizes from sector information, identifying the boot file system used in the linux/window operating system and also understanding the windows boot process.

Lab Objectives

The lab is to understand some PC system fundamentals and make some system partition calculation.

Tools and Resources Used

Window 11/pc

Linux/kali vm

Calculator

Methology

I navigate to the windows command line interface where I execute different command.

Screenshot and Evidence

PART A:- System Identification and Evidence Layer Mapping

```

PS C:\Users\PC> Get-Disk

Number Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
-----
0 TOSHIBA MQ... 12CEPI03T Healthy Online 931.51 GB GPT

PS C:\Users\PC> Get-Partition

DiskPath: \\?\scsi#disk&ven_ata&prod_toshiba_mq04abf1#4&3b702f65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber DriveLetter Offset Size Type
-----
1 1048576 100 MB System
2 105906176 16 MB Reserved
3 122683392 930.61 GB Basic
4 999352696832 811 MB Recovery

PS C:\Users\PC> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
-----
C NTFS NTFS Fixed Healthy OK 621.86 GB 930.61 GB
C NTFS NTFS Fixed Healthy OK 73.52 MB 811 MB

PS C:\Users\PC> |

```

SYSTEM LAYERS	TYPE OF EVIDENCE THAT CAN BE GENERATED
User layer	Users frequent activities such as the website he visit always, the folders he is always in touch with.
Application layer	The meta data of the application the user is always interacting with
Operating system layer	Users activity logs which can be found in the window event logs.
Disk layer	The files installed and deleted on the users disk
Memory/process layer	The PID of running processes

Question

1. Explain why evidence may exist in more than one layer.

Answer:-

The reason why evidence may exist in more than layer is because when a user start up an application other activity start coming such as the PID of that application the user start up and also the event log also keep record about that application that the user is running.

PART B:- PROGRAM, PROCESS AND THREAD INVESTIGATION

```

C:\Windows\system32\cmd.e  X  +  v

C:\Users\PC>tasklist

Image Name                      PID Session Name        Session#    Mem Usage
=====
System Idle Process             0 Services              0            8 K
System                          4 Services              0          11,476 K
Secure System                   140 Services             0          47,760 K
Registry                       184 Services             0          30,608 K
smss.exe                       744 Services              0           1,592 K
csrss.exe                      940 Services              0           6,496 K
csrss.exe                     1100 Console                1           8,056 K
wininit.exe                    1148 Services             0           9,256 K
winlogon.exe                   1168 Console               1          16,448 K
services.exe                   1224 Services              0          16,248 K
LsaIso.exe                     1268 Services             0           4,092 K
lsass.exe                      1276 Services             0          30,908 K
svchost.exe                    1428 Services             0          43,848 K
WUDFHost.exe                   1460 Services             0          22,640 K
fontdrvhost.exe               1496 Services             0           4,100 K
fontdrvhost.exe               1504 Console               1          11,368 K
svchost.exe                    1592 Services             0          22,708 K
svchost.exe                    1640 Services             0          10,988 K
WUDFHost.exe                   1708 Services             0          10,348 K
dwm.exe                        1804 Console               1          110,296 K
svchost.exe                    1924 Services             0          10,856 K
svchost.exe                    1932 Services             0           6,948 K
svchost.exe                    1964 Services             0          11,316 K
svchost.exe                     936 Services             0          12,388 K
svchost.exe                     724 Services             0          10,576 K
svchost.exe                    1000 Services             0          17,604 K
svchost.exe                    1328 Services             0          14,512 K
svchost.exe                    1304 Services             0          18,852 K
svchost.exe                    2084 Services             0          11,028 K
svchost.exe                    2100 Services             0          15,396 K

```

PROCESS	PID VALUES
svchost.exe	2100
WUDFHOST.exe	1804
smss.exe	744

Question

1. In your report, explain the difference between a program, a process, and a thread in your own words.

Answer

Program	Process	Thread
A program is a set of instructions is used to tell a computer how to perform a particular task.	A process is simply a program that is in execution	A thread can be defined as the smallest unit of execution within a process

- Explain why process information may be useful during malware analysis or live forensic triage.

Answer:-the reason why process information may be useful during malware analysis or live forensic triage is that it helps in providing some basic important information about a program/software/application or service that were running before the occurrence of an incident(malware attack)

PART C:- DISK SECTOR AND DISK SIZE CALCULATION

```
DISKPART> list partition
```

Partition ###	Type	Size	Offset
Partition 1	System	100 MB	1024 KB
Partition 2	Reserved	16 MB	101 MB
Partition 3	Primary	930 GB	117 MB
Partition 4	Recovery	811 MB	930 GB

```
DISKPART> _
```

The formular I will be using include

Start sector = Starting Offset ÷ 512

Number of sector = Partition Size ÷ 512

A) EFI/SYSTEM PARTITION

Start sector = 1,048,576 ÷ 512 = 2,048

Number of sector = 104,857,600 ÷ 512 = 204,800 sectors

Size = 104,857,600 ÷ 1,048,576 = 100 MB

B) C: PARTITION

Start sector = 122,683,392 ÷ 512 = 239,616

Number of sector = 999,230,013,440 ÷ 512 = 1,951,621,120 sectors

Size = 999,230,013,440 ÷ 1,073,741,824 = 930.61 GB

PART D:- SAFE PARTITION TABLE INSPECTION

```
PS C:\Users\PC> Get-Disk

Number Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
-----
0 TOSHIBA MQ... 12CEPI03T Healthy Online 931.51 GB GPT

PS C:\Users\PC> Get-Partition

DiskPath: \\?\scsi#disk&ven_ata&prod_toshiba_mq04abf1#4&3b702f65&0&020000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber DriveLetter Offset Size Type
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PS C:\Users\PC> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
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C NTFS Fixed Healthy OK 621.86 GB 930.61 GB
C NTFS Fixed Healthy OK 73.52 MB 811 MB

PS C:\Users\PC> |
```

```
C:\Windows\system32\diskpart.exe

Microsoft DiskPart version 10.0.26100.1150
Copyright (C) Microsoft Corporation.
On computer: MUNTAKA

DISKPART> list disk

Disk ### Status Size Free Dyn Gpt
-----
Disk 0 Online 931 GB 1024 KB *

DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

Partition ### Type Size Offset
-----
Partition 1 System 100 MB 1024 KB
Partition 2 Reserved 16 MB 101 MB
Partition 3 Primary 930 GB 117 MB
Partition 4 Recovery 811 MB 930 GB

DISKPART> _
```

```

(kali@kali)~$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME        SIZE TYPE FSTYPE MOUNTPOINT
sda          80.1G disk
└─sda1 80.1G part ext4    /

(kali@kali)~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSE% MOUNTPOINTS
└─sda1 ext4   1.0    root  832cb7bc-501f-4fd2-8a6e-28e72fbccc16   54G     26% /

(kali@kali)~$ blkid
zsh: bad pattern: ^[[200~df

(kali@kali)~$ df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
udev            devtmpfs  917M   0    917M   0% /dev
tmpfs           tmpfs     197M   1.3M 196M   1% /run
/dev/sda1       ext4      79G   21G   55G   28% /
tmpfs           tmpfs     983M   4.0K 983M   1% /dev/shm
tmpfs           tmpfs     5.0M   0    5.0M   0% /run/lock
tmpfs           tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     983M   8.0K 983M   1% /tmp
tmpfs           tmpfs     1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197M   124K 197M   1% /run/user/1000

```

PART E:- MBR, GPT and Boot Process Explanation

The list below explain how a pc boots

- After pressing the power on button the following occur
- The computer conduct a power on self test to check and confirm that the hardware are OK
- The basic input out system applied the firmware setting
- The GBT boot locate the bootable partition
- The boot loader/manager select the available operating system and if there is an alternative a user is prompted to choose before the pre kernel loading occurs.
- The kernel and operating system control take control at this step whereby the operating system start up fully and then prompt a user to enter his password in order to have access to his system.

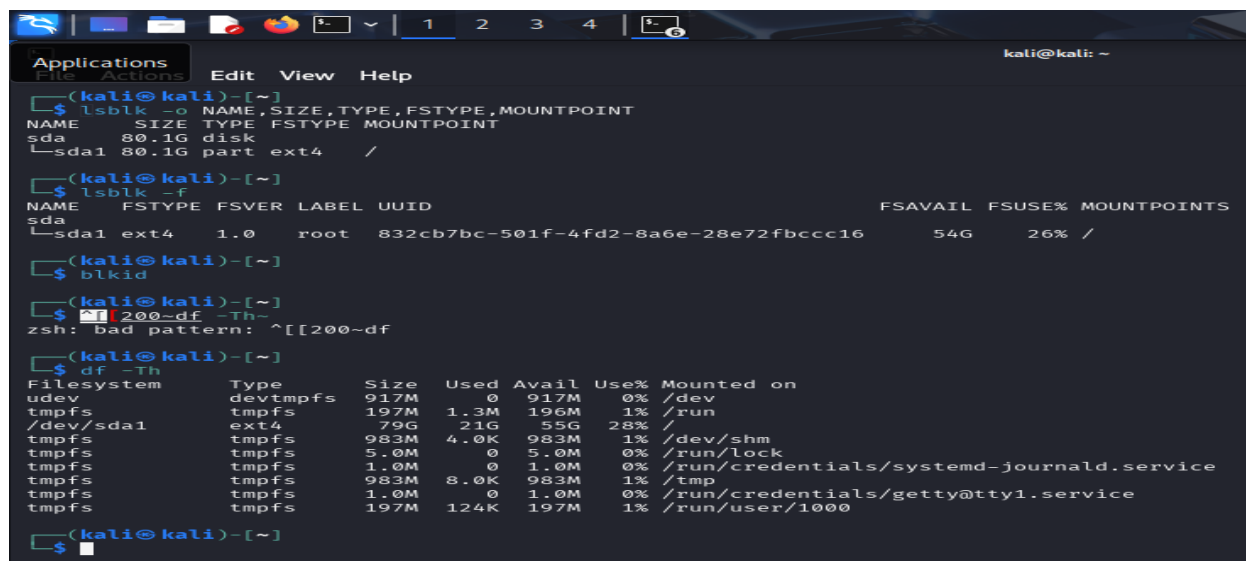
PART F:- File System Identification and Interpretation

```

PS C:\Users\PC> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
-----
C            NTFS           NTFS           Fixed      Healthy      OK              621.86 GB 930.61 GB
C            NTFS           NTFS           Fixed      Healthy      OK              73.52 MB 811 MB

```



```
(kali@kali)~$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME        SIZE TYPE FSTYPE MOUNTPOINT
sda          80.1G disk
└─sda1 80.1G part ext4 /

(kali@kali)~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
┌─sda1 ext4   1.0   root  832cb7bc-501f-4fd2-8a6e-28e72fbccc16   54G    26% /

(kali@kali)~$ blkid
zsh: bad pattern: ^[[200~df

(kali@kali)~$ df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
udev            devtmpfs  917M   0    917M   0% /dev
tmpfs           tmpfs     197M   1.3M 196M   1% /run
/dev/sda1       ext4      79G   21G   55G   28% /
tmpfs           tmpfs     983M   4.0K 983M   1% /dev/shm
tmpfs           tmpfs     5.0M   0    5.0M   0% /run/lock
tmpfs           tmpfs     1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           tmpfs     983M   8.0K 983M   1% /tmp
tmpfs           tmpfs     1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           tmpfs     197M   124K 197M   1% /run/user/1000
```

Question

1. Identify the file system used by at least two volumes or partitions on your system.

Answer:- the file that were used by two partition on my system are the EFI and NTFS filesystem

2. For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.

Answer:- the window system uses NTFS file system

3. For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.

Answer:- the linux system uses ext4 filesystem

Conclusion

In conclusion the lab has expose me to some of the window system findamentals that I don't know before and also having a strong understanding about window partition and the file system

Reference

- ICDFA course Material (BVDF201(basic computer skills for digital forensics)- by COMMANDANT AMINU IDRIS
 - ICDFA Recorded sessions
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